CACTUS FALL 2026 · WEEK 5 · LEARNING SESSION

Analyze text

Turning text into numbers loses things. Today you learn what each step keeps and what it hides.

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THE IDEA

Text is not numbers

Text

Words, names, jokes, scripts, emoji, and context.

Numbers

Counts, scores, and groups that a computer can add up.

In between

Every step makes a choice: where a word ends, what to drop, what a word is worth.

Today: tokens, counts, dictionary sentiment, and clusters. For each one, ask: **what did this step hide?**

WHAT IS A TOKEN?

The unit you count

Words

Split on spaces. Works for English. Punctuation sticks to words.

Characters

Every character is a token. Works for anything, and means little.

Subwords

Pieces of words ("un" + "happy"). What large language models use.

The rule you choose changes every number that comes after it.
Chinese is written **without spaces between words**, so splitting on spaces does not work.

LIVE DEMO

One sentence, different tokenizers

English	AI isn't replacing teachers, it's changing homework. SPACES (7) `AI` `isn't` `replacing` `teachers,` `it's` `changing` `homework.` JIEBA (13) `AI` `isn` `'` `t` `replacing` `teachers` `,` `it` `'` `s` `changing` `homework` `.`
Chinese	新加坡推出人工智能學習計劃 SPACES (1) `新加坡推出人工智能學習計劃` JIEBA (5) `新加坡` `推出` `人工智能` `學習` `計劃`
Mixed	Singapore 學生 use ChatGPT 寫功課 SPACES (5) `Singapore` `學生` `use` `ChatGPT` `寫功課` JIEBA (6) `Singapore` `學生` `use` `ChatGPT` `寫功` `課`

jieba splits Chinese into words, but it is not perfect: it cut 寫功課 into 寫功 + 課. It also splits English punctuation. No tokenizer is right for every text.

COUNTING

Raw counts mislead

Raw top words

the, in, for, of, and … The little words win in every corpus. They tell you about English, not your topic.

Cleaning steps

Lowercase. Remove stopwords. Drop numbers or links. Each is a choice.

What can go wrong

Lowercasing turns the country "US" into "us". "us" is a stopword. Dropping short words deletes "AI".

Always read the top words after cleaning. Ask what was deleted that you wanted.

DISTINCTIVE TERMS

What makes this group different?

A word	Share in group A	Share in group B
the	5.1%	5.0%
tutor	**0.9%**	**0.1%**
of	2.4%	2.5%

Compare a word's share in one group with its share in the others. "the" is common but not distinctive. "tutor" is rarer but tells the groups apart. (Invented numbers.)

Weighting methods such as TF-IDF and log ratios do this for you. They still depend on the cleaning choices before them.

DICTIONARY SENTIMENT

Add up the word scores

```
great = +3   good = +2   bad = -2

"The service was bad" → -2 → negative
"This is not good" → +2 → positive?
    with a rule: after "not", flip the next word → -2 → negative
```

A **dictionary** gives each word a score and adds them up. VADER is an English dictionary made for social media, with rules for negation, capitals, and "!". It gives a compound score from -1 to +1.

WHERE IT FAILS

A small gallery

Sarcasm

"Oh great, it deleted my file again." The dictionary sees "great".

A name

"Joy asked about the results." Joy is a person here.

Slang

"That demo was sick!" Positive for a person, negative in the dictionary.

Mixed feelings

"Great team, terrible pay." Plus and minus add up to nothing, and that hides both.

Long texts

Scores tend to go to the extremes in long posts, so averages of long and short texts do not compare well.

What to do

Read a sample by hand. Report how often you and the dictionary disagree.

READ BEFORE YOU NAME

Top words are a summary, not the evidence

The method

A clustering method groups texts that use similar words, and prints each group's top words.

The trap

A label is easy to invent from five words. The examples may say something else.

Coherent

Five examples, one topic: you can name it.

Too mixed

General words, unrelated examples: do not name it.

Boilerplate

"Subscribe", "sign in", "newsletter": not content. Remove it.

Read three to five examples from each cluster before you name it.

WHAT TO DO NEXT

Before the coding session

Practice page

What Counts as a Word?, about 20 minutes. It ends with a tokenizing stub for Colab.

Required reading

Textbook Chapter 10, sections 10.1 and 10.4, and Chapter 11, section 11.3.

Data

The class corpus is on the class drive. Do not post it or paste it into an AI chat.

In the coding session: tokenize, count, score sentiment, read ten texts by hand, and find three mistakes in AI-written code.

QUESTIONS

What will your method hide?

Think about your own project's text. Which of today's steps would change your numbers the most?